Supervyse OPF Intel Test Utility User Guide

Revision 1.0

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2023-11-09	Roger Chen	Edwin Liu	Initial version
1.1	2024-9-9	Nick Lin	Austin Su	

1 DIMM Devices Accessor

1.1 Introduction

DIMM Devices Accessor is the tool provided by **Intel** (proprietary, not released for public at Github) which can be used to check any information in the DIMM memory. DIMM Devices Accessor doesn't explicitly check the type of CPU it's communicating with. It should work on SPR, GNR, and similar CPUs. For more information, please refer to 1.4 Reference.

1.2 Usage

At BMC side, issue the following command to verify DIMM device depends on the input json file, e.g. input.json. **<cpu address>**, **<imc>**, **<channel>**, **<slot>** are parameters to identify the location of DIMM device.

```
# dimm-devices-accessor-example <cpu address> <imc> <channel> <slot> <input.json>
```

<cpu address>: The address of CPU that DIMM Device Accessor is communicating to.

	CPU0	CPU1	CPU2	CPU3	CPU4	CPU5	CPU6	CPU7
CPU Address	0x30	0x31	0x32	0x33	0x34	0x35	0x36	0x37

<input.json>

```
{
    "SpdHub_VM":
    {
        "Write":
        [
            {"Register": "0x1a", "Values": ["0x1"]}
        ],
        "Read":
        [
            {"Register": "0x1c", "Length": "0x1"}
        ]
    },
    "SpdHub_NVM":
    {
        "Write":
        [
            {"Register": "0x301", "Values": ["0xc1", "0x0f", "0x02", "0xaf"]}
        ],
        "Read":
        [

        ]
    },
      "PMIC_HR":
    {
        "Write":
        [
            {"Register": "0x30", "Values": ["0x1"]}
        ],
        "Read":
        [
            {"Register": "0xc", "Length": "0x1"},
            {"Register": "0xd", "Length": "0x1"},
            {"Register": "0xe", "Length": "0x1"},
            {"Register": "0xf", "Length": "0x1"}
        ]
    },
    "PMIC_VR":
    {
        "Write":
        [

        ],
        "Read":
        [
            {"Register": "0x4e", "Length": "0x1"}
        ]
    },
    "RCD_Paged":
    {
        "Write":
        [
            {"Register": "0x60", "Values": ["0x04"]}
        ],
        "Read":
        [

        ]
    },
    "RCD_Unpaged":
    {
        "Write":
        [
            {"Register": "0x12", "Values": ["0x1"]}
        ],
        "Read":
        [
            {"Register": "0x5", "Length": "0x1"}
        ]
    }
}
}
```

Node	Description
SpdHub_VM	SPD Hub Volatile Memory

SpdHub_NVM	SPD Hub Non-Volatile Memory
PMIC_HR	PMIC Host Region
PMIC_VR	PMIC Vendor Region
RCD_Paged	RCD Paged Control Word(s)
RCD_Unpaged	RCD Unpaged Control Word(s)

For each node, there are 2 operations. **Read** and **Write**.

Here is the example for Read/Write operations, with parameters **Register**, **Values** and **Length**.

- Register: The offset address of node device for Read/Write operations.
- Values: An array or a single byte for **Write** operation.
- Length: The length for **Read** operation.

```
"SpdHub_VM":
{
    "Write":
    [
        {"Register": "0x1a", "Values": ["0x1"]}
    ],
    "Read":
    [
        {"Register": "0x1c", "Length": "0x1"}
    ]
},
```

For more info, please refer to **/var/lib/dda-example**, e.g. *dda_test_config.json.*

1.3 Example

Here is an example to show how to use. e.g. ***dimm-devices-accessor-example 48 0 1 0***

- <cpu address>: 48 in decimal, which represent the address of CPU.
- <imc>: 0, the index of imc controller.
- <channel>: 1, the index of channel.
- <slot>: 0, the index of slot.
- <input.json>: Not mentioned, which means the default json file will be loaded.

```
$ dimm-devices-accessor-example 48 0 1 0

~~Dimm Devices OOB Accessor example~~
Initializing accessor for CpuAddress=48 Imc=0 Channel=1 Slot=0
---------------------------------
SPD Hub - Volatile Memory test start
```

```
----------------------------------
Started READ Volatile Memory. Address=0x1a Length=1
Finished READ Volatile Memory. Address=0x1a Length=1
Result = 0x00
Started WRITE Volatile Memory. Address=0x1a Data=0x01
Finished WRITE Volatile Memory. Address=0x1a Data=0x01
Started READ Volatile Memory. Address=0x1a Length=1
Finished READ Volatile Memory. Address=0x1a Length=1
Result = 0x01
----------------------------------
SPD Hub - Non-Volatile Memory test start
----------------------------------
Started READ Non-Volatile Memory. Address=0x301 Length=4
Finished READ Non-Volatile Memory. Address=0x301 Length=4
Result = 0x00 00 00 00
Started WRITE Non-Volatile Memory. Address=0x301 Data=0xc1 0f 02 af
Finished WRITE Non-Volatile Memory. Address=0x301 Data=0xc1 0f 02 af
Started READ Non-Volatile Memory. Address=0x301 Length=4
Finished READ Non-Volatile Memory. Address=0x301 Length=4
Result = 0xc1 0f 02 af
~~Using PMIC ID 1001 as default (assuming PID pin connected to GND on PCB)~~
----------------------------------
PMIC - Host Region test start
----------------------------------
Started READ Host Region. Address=0x30 Length=1
Finished READ Host Region. Address=0x30 Length=1
Result = 0x00
Started WRITE Host Region. Address=0x30 Data=0x01
Finished WRITE Host Region. Address=0x30 Data=0x01
Started READ Host Region. Address=0x30 Length=1
Finished READ Host Region. Address=0x30 Length=1
Result = 0x01
----------------------------------
PMIC - DIMM Vendor Region test start
----------------------------------
Started UNLOCKING DIMM Vendor Region. Password=0x9473
Finished UNLOCKING DIMM Vendor Region. Password=0x9473
Started READ DIMM Vendor Region. Address=0x4e Length=1
Finished READ DIMM Vendor Region. Address=0x4e Length=1
Result = 0x99
Started LOCKING DIMM Vendor Region
Finished LOCKING DIMM Vendor Region
----------------------------------
RCD - Unpaged Control Words (SidebandBus) test start
----------------------------------
Started READ Unpaged Control Word(s). Address=0x12 Channel=0 Length=1
Finished READ Unpaged Control Word(s). Address=0x12 Channel=0 Length=1
Result = 0x00
Started WRITE Unpaged Control Word(s). Address=0x12 Channel=0 Data=0x01
Finished WRITE Unpaged Control Word(s). Address=0x12 Channel=0 Data=0x01
Started READ Unpaged Control Word(s). Address=0x12 Channel=0 Length=1
Finished READ Unpaged Control Word(s). Address=0x12 Channel=0 Length=1
Result = 0x01
----------------------------------
RCD - Paged Control Words (SidebandBus) test start
----------------------------------
Started READ Paged Control Word(s). Address=0x60 Page=1 Channel=1 Length=1
Finished READ Paged Control Word(s). Address=0x60 Page=1 Channel=1 Length=1
Result = 0x00
Started WRITE Paged Control Word(s).Address=0x60 Page=1 Channel=1 Data=0x04
Finished WRITE Paged Control Word(s). Address=0x60 Page=1 Channel=1 Data=0x04
```

```
Started READ Paged Control Word(s). Address=0x60 Page=1 Channel=1 Length=1
Finished READ Paged Control Word(s). Address=0x60 Page=1 Channel=1 Length=1
Result = 0x04
~~~~Tests finished~~~~
~~~~Cleaning up registers to default states~~~~
Cleaned up SPD Hub example registers
Cleaned up PMIC example registers
Cleaned up RCD example registers
~~~~Cleaning up finished~~~~
```

1.4 Reference

- 794030_Intel_BHS_DIMM_Devices_Accessor_Design_Rev0_95
- SPD Semaphore Rev. 1.0: https://cdrdv2.intel.com/v1/dl/getContent/705375
- **JESD300-5**: SPD5118, SPD5108 Hub and Serial Presence Detect Device Standard
- **JESD400-5**: Proposed DDR5 Serial Presence Detect (SPD) Contents
- **JESD301-1**: PMIC50x0 Power Management IC Specification, Rev 1.0
- MIPI I3C HCI version 1.2
- Granite Rapids CPU EDS vol1 (**RDC#639866**) and vol2 (**RDC#637776**)
- Sapphire Rapids and Emerald Rapids Serial Presence Detect (SPD) Bus Recovery and Throttling: Technical Advisory (**RDC#727124**)